

Integrated Controls System

Data from a real \$1B LNG terminal project (anonymized)

THE PROBLEM

- Mega-projects run on 5+ disconnected systems — planning, cost, engineering, procurement, field progress.
- Teams spend hours each cycle manually reconciling numbers; risk of errors grows as data volumes increase.
- Conflicts between systems are caught at the final report stage — not at the source.

WHAT I BUILT

An integration layer that consolidates data from existing project systems into one decision interface.

- **Clear reporting confidence** — one consolidated view, no reconciliation debates
- **Reduces manual cross-checking** — automates basic checks teams currently do by hand
- **Better data quality** — flags staleness, checks completeness, surfaces contradictions
- **Reduces reconciliation time** — less time stitching spreadsheets, more time on decisions
- **No server, no vendor lock-in** — SSOT spreadsheet → Python ETL → JSON → Chart.js → GitHub Pages

SEE IT Live dashboard (prototype — same logic applies in Power BI or enterprise stack):

See it live → <https://mamdohabdu.github.io/lng-dashboard>

WHAT THIS DEMONSTRATES

- Cross-system data integration across 5+ sources
- Automated contradiction detection and staleness monitoring
- Schedule analytics: float erosion, productivity trending, delay attribution
- Recovery plan validation against historical performance
- Consolidated view that supports decision-making
- Flags inconsistencies for review — does not override source systems